Import Data

In [57]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

In [58]:
```
data = pd.read_csv(r"D:\Downloads\Udemy-PY\Spotify_Youtube Dataset.csv")
```

In [59]:
```
data.head()
```

Out[59]:

	Unnamed: 0	Artist	Url_spotify	Track	Album
0	0	Gorillaz	https://open.spotify.com/artist/3AA28KZvwAUcZu...	Feel Good Inc.	Demon Days
1	1	Gorillaz	https://open.spotify.com/artist/3AA28KZvwAUcZu...	Rhinestone Eyes	Plastic Beach
2	2	Gorillaz	https://open.spotify.com/artist/3AA28KZvwAUcZu...	New Gold (feat. Tame Impala and Bootie Brown)	New Gold (feat. Tame Impala and Bootie Brown)
3	3	Gorillaz	https://open.spotify.com/artist/3AA28KZvwAUcZu...	On Melancholy Hill	Plastic Beach
4	4	Gorillaz	https://open.spotify.com/artist/3AA28KZvwAUcZu...	Clint Eastwood	Gorillaz

5 rows × 28 columns

In [60]:
```
data.columns
```

Out[60]:
```
Index(['Unnamed: 0', 'Artist', 'Url_spotify', 'Track', 'Album', 'Album_type',
       'Uri', 'Danceability', 'Energy', 'Key', 'Loudness', 'Speechiness',
       'Acousticness', 'Instrumentalness', 'Liveness', 'Valence', 'Tempo',
       'Duration_ms', 'Url_youtube', 'Title', 'Channel', 'Views', 'Likes',
       'Comments', 'Description', 'Licensed', 'official_video', 'Stream'],
      dtype='object')
```

In [61]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20718 entries, 0 to 20717
Data columns (total 28 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Unnamed: 0        20718 non-null  int64  
 1   Artist            20718 non-null  object 
 2   Url_spotify       20718 non-null  object 
 3   Track             20718 non-null  object 
 4   Album             20718 non-null  object 
 5   Album_type        20718 non-null  object 
 6   Uri               20718 non-null  object 
 7   Danceability      20716 non-null  float64
 8   Energy            20716 non-null  float64
 9   Key               20716 non-null  float64
 10  Loudness          20716 non-null  float64
 11  Speechiness       20716 non-null  float64
 12  Acousticness      20716 non-null  float64
 13  Instrumentalness  20716 non-null  float64
 14  Liveness          20716 non-null  float64
 15  Valence           20716 non-null  float64
 16  Tempo             20716 non-null  float64
 17  Duration_ms       20716 non-null  float64
 18  Url_youtube       20248 non-null  object 
 19  Title             20248 non-null  object 
 20  Channel           20248 non-null  object 
 21  Views             20248 non-null  float64
 22  Likes             20177 non-null  float64
 23  Comments          20149 non-null  float64
 24  Description       19842 non-null  object 
 25  Licensed          20248 non-null  object 
 26  official_video    20248 non-null  object 
 27  Stream            20142 non-null  float64
dtypes: float64(15), int64(1), object(12)
memory usage: 4.4+ MB
```

In [62]:
```
data.drop(columns = ['Unnamed: 0','Uri','Url_spotify','Url_youtube'], inplace = Tru
```

In [63]:
```
data.isna().sum()
```

Out[63]:
```
Artist                0
Track                 0
Album                 0
Album_type            0
Danceability          2
Energy                2
Key                   2
Loudness              2
Speechiness           2
Acousticness          2
Instrumentalness      2
Liveness              2
Valence               2
Tempo                 2
Duration_ms           2
Title               470
Channel             470
Views               470
Likes               541
Comments            569
Description         876
Licensed            470
official_video      470
Stream              576
dtype: int64
```

In [64]:
```
data['Likes'] = data['Likes'].fillna(0)
data['Comments'] = data['Comments'].fillna(0)
```

In [65]:
```
data.dropna(inplace = True) # Removed all missing values
```

1. Show top 10 Artists with highest views from YouTube

In [66]:
```
Artists_grouped = data.groupby('Artist')['Views'].sum()
Artists_grouped
```

Out[66]:
```
Artist
$NOT              1.107849e+08
$uicideboy$       3.317202e+08
(G)I-DLE          1.754954e+09
*NSYNC            1.027833e+09
070 Shake         9.609936e+07
                      ...
will.i.am         2.830801e+09
Ángela Aguilar    1.385295e+09
Ñejo              6.266808e+08
Ñengo Flow        8.127263e+08
Øneheart          3.462331e+07
Name: Views, Length: 2040, dtype: float64
```

In [67]:
```
Artists_grouped.sort_values(ascending = False).head(10)
```

Out[67]:
```
Artist
Ed Sheeran                  1.546021e+10
CoComelon                   1.460167e+10
Katy Perry                  1.312063e+10
Charlie Puth                1.216759e+10
Luis Fonsi                  1.162811e+10
Justin Bieber               1.099079e+10
Daddy Yankee                1.086828e+10
Bruno Mars                  1.023184e+10
Macklemore & Ryan Lewis     1.012206e+10
Coldplay                    9.997278e+09
Name: Views, dtype: float64
```

2. Top 10 Tracks with highest stream on Spotify

In [68]:
```
data.head(2)
```

Out[68]:

	Artist	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speechi
0	Gorillaz	Feel Good Inc.	Demon Days	album	0.818	0.705	6.0	-6.679	0.:
1	Gorillaz	Rhinestone Eyes	Plastic Beach	album	0.676	0.703	8.0	-5.815	0.(

2 rows × 24 columns

In [69]:
```
Track_streams = data.groupby('Track')['Stream'].sum()
Track_streams
```

Out[69]:
```
Track
!ly (feat. Coez)                52628204.0
#1 - Colby O'Donis Remix          561794.0
#3                              45002482.0
#41                             34528391.0
#Ysya2020 Vol. 5 - Silbando     26297019.0
                                    ...
高嶺の花子さん                    69373615.0
그대라는 시                        84972553.0
눈,코,입 (Eyes, Nose, Lips)       112300290.0
러시안 룰렛 Russian Roulette       151307068.0
링가링가 (RINGA LINGA)             36450721.0
Name: Stream, Length: 16662, dtype: float64
```

In [70]:
```
Track_streams.sort_values(ascending = False).head(10)
```

Out[70]:
```
Track
Closer                                                          5.465365e+09
Can't Hold Us (feat. Ray Dalton)                                5.225629e+09
Sunflower - Spider-Man: Into the Spider-Verse                   5.076660e+09
Happier                                                         4.757615e+09
STAY (with Justin Bieber)                                       4.731555e+09
Señorita                                                        4.672440e+09
The Middle                                                      4.566883e+09
Eastside (with Halsey & Khalid)                                 4.274888e+09
lovely (with Khalid)                                            4.221148e+09
Enemy (with JID) - from the series Arcane League of Legends    4.182328e+09
Name: Stream, dtype: float64
```

In [71]:
```python
x = data[['Track', 'Stream']]          # By different way
x
```

Out[71]:

	Track	Stream
0	Feel Good Inc.	1.040235e+09
1	Rhinestone Eyes	3.100837e+08
2	New Gold (feat. Tame Impala and Bootie Brown)	6.306347e+07
3	On Melancholy Hill	4.346636e+08
4	Clint Eastwood	6.172597e+08
...	...	...
20713	JUST DANCE HARDSTYLE	9.227144e+06
20714	SET FIRE TO THE RAIN HARDSTYLE	1.089818e+07
20715	OUTSIDE HARDSTYLE SPED UP	6.226110e+06
20716	ONLY GIRL HARDSTYLE	6.873961e+06
20717	MISS YOU HARDSTYLE	5.695584e+06

19298 rows × 2 columns

In [72]:
```python
x.sort_values(by = 'Stream', ascending = False ).head(10)
```

Out[72]:

	Track	Stream
15250	Blinding Lights	3.386520e+09
12452	Shape of You	3.362005e+09
19186	Someone You Loved	2.634013e+09
17937	rockstar (feat. 21 Savage)	2.594927e+09
17445	Sunflower - Spider-Man: Into the Spider-Verse	2.538330e+09
17938	Sunflower - Spider-Man: Into the Spider-Verse	2.538330e+09
13503	One Dance	2.522432e+09
16099	Closer	2.456205e+09
16028	Closer	2.456205e+09
14030	Believer	2.369272e+09

3. What are the most common Album on Spotify?

In [73]:
```
data['Album_type'].unique()
```

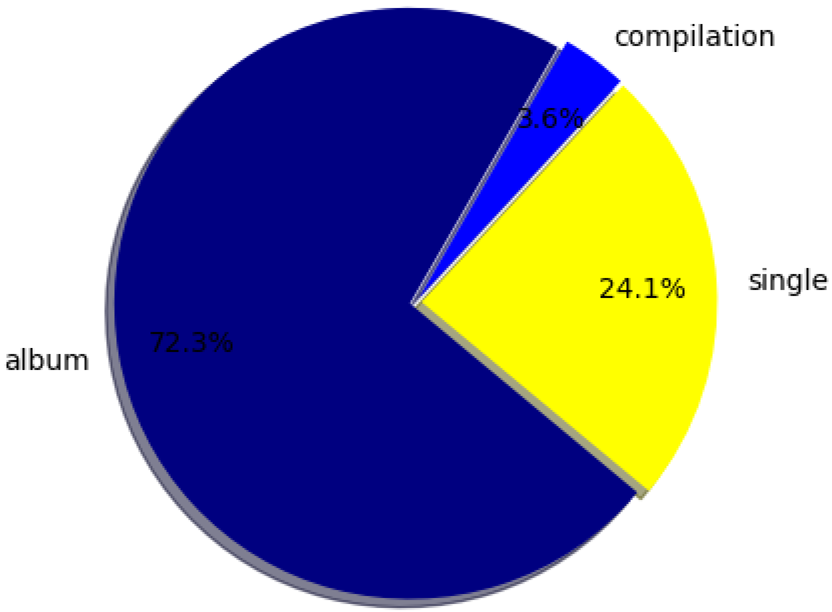
Out[73]:
```
array(['album', 'single', 'compilation'], dtype=object)
```

In [74]:
```
c = data['Album_type'].value_counts()
c
```

Out[74]:
```
Album_type
album          13952
single          4653
compilation      693
Name: count, dtype: int64
```

In [75]:
```
plt.pie(c, labels = c.index, autopct = "%1.1f%%", startangle = 60, colors = ['navy'
plt.show
```

Out[75]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



4. How do different Comments, Likes and Views are compared between different Album types?

In [76]:
```
df = data.groupby('Album_type')[['Views','Likes','Comments']].mean()
df
```

Out[76]:

Album_type	Views	Likes	Comments
album	1.014359e+08	672593.609948	29032.951548
compilation	8.500535e+07	544275.818182	18074.903319
single	8.480843e+07	722266.303890	27313.048571

In [77]:
```
df = df.reset_index()
df
```

Out[77]:

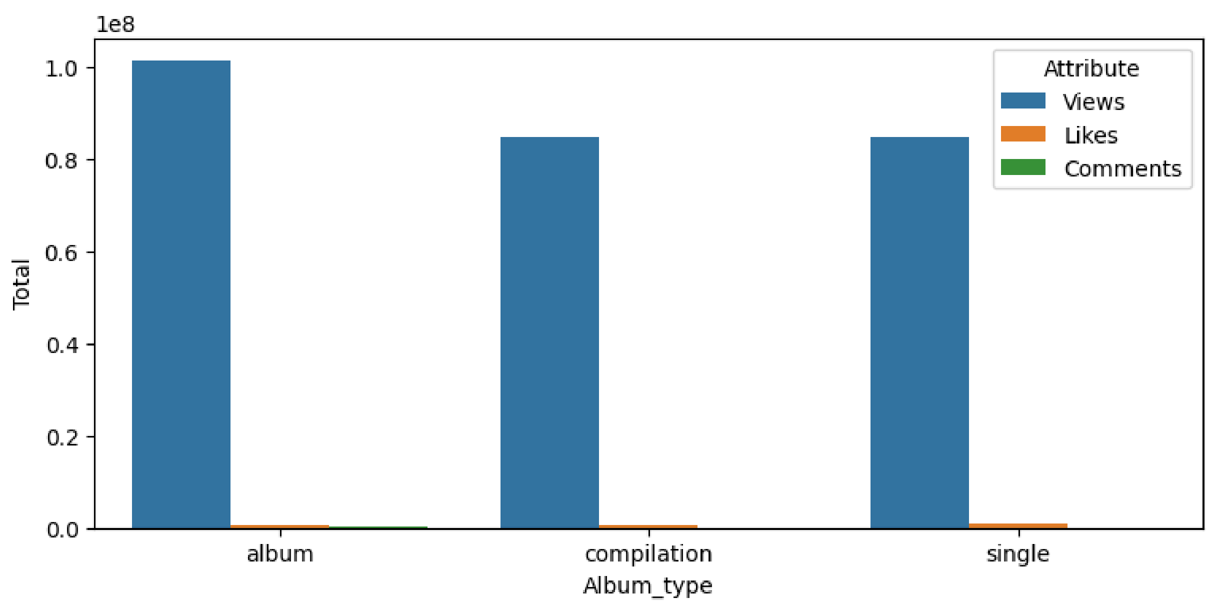
	Album_type	Views	Likes	Comments
0	album	1.014359e+08	672593.609948	29032.951548
1	compilation	8.500535e+07	544275.818182	18074.903319
2	single	8.480843e+07	722266.303890	27313.048571

In [78]:
```
df_reshape = pd.melt(df, id_vars = 'Album_type', var_name = 'Attribute', value_name
df_reshape
```

Out[78]:

	Album_type	Attribute	Total
0	album	Views	1.014359e+08
1	compilation	Views	8.500535e+07
2	single	Views	8.480843e+07
3	album	Likes	6.725936e+05
4	compilation	Likes	5.442758e+05
5	single	Likes	7.222663e+05
6	album	Comments	2.903295e+04
7	compilation	Comments	1.807490e+04
8	single	Comments	2.731305e+04

In [79]:
```
plt.figure(figsize=(9,4))
sns.barplot(x = 'Album_type', y = 'Total',hue = 'Attribute', data = df_reshape
           );
```



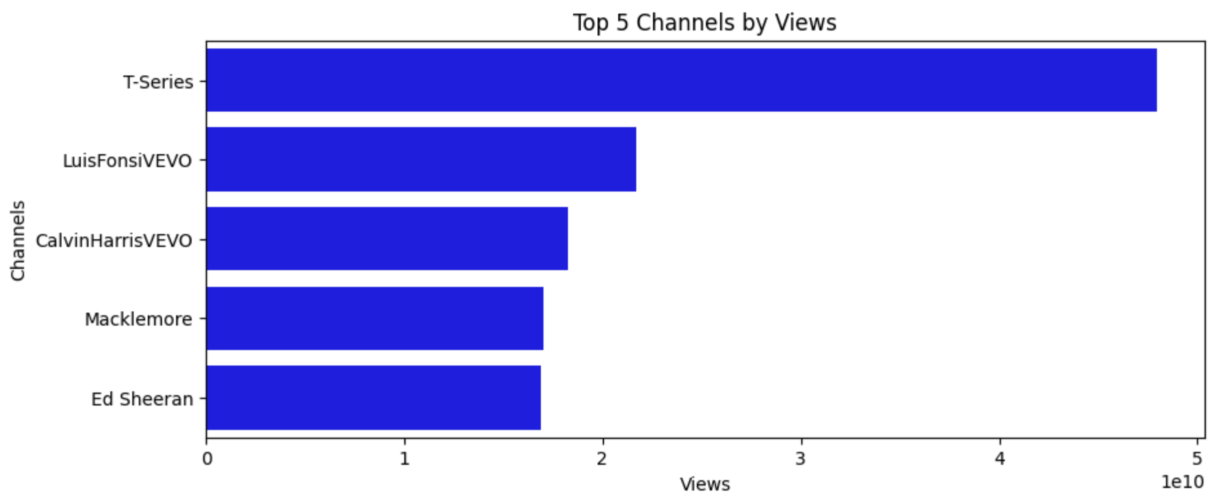
5. Show top 5 Youtube channels on the base of Views

In [83]:
```
yt_views = data.groupby('Channel')['Views'].sum().sort_values(ascending = False).he
yt_views
```

Out[83]:
```
Channel
T-Series            4.796159e+10
LuisFonsiVEVO       2.170252e+10
CalvinHarrisVEVO    1.828114e+10
Macklemore          1.700341e+10
Ed Sheeran          1.691822e+10
Name: Views, dtype: float64
```

In [92]:
```
yt_views = yt_views.reset_index()
```

In [101…
```
plt.figure(figsize = (10,4))
sns.barplot(x = 'Views', y = 'Channel', data = yt_views, color = 'blue' )
plt.title('Top 5 Channels by Views')
plt.xlabel('Views')
plt.ylabel('Channels');
```



6. Show the top most Track based on views

In [104…
```
data.sort_values(by = 'Views', ascending = False).head(1)
```

Out[104…

	Artist	Track	Album	Album_type	Danceability	Energy	Key	Loudness	Speech
1147	Luis Fonsi	Despacito	VIDA	album	0.655	0.797	2.0	-4.787	

1 rows × 24 columns

7. Which toop 7 Tracks have highest Like to Views ratio?

In [106…
```
track_lv = data[['Track', 'Views', 'Likes']]
track_lv
```

Out[106…

	Track	Views	Likes
0	Feel Good Inc.	693555221.0	6220896.0
1	Rhinestone Eyes	72011645.0	1079128.0
2	New Gold (feat. Tame Impala and Bootie Brown)	8435055.0	282142.0
3	On Melancholy Hill	211754952.0	1788577.0
4	Clint Eastwood	618480958.0	6197318.0
...	...	...	...
20713	JUST DANCE HARDSTYLE	71678.0	1113.0
20714	SET FIRE TO THE RAIN HARDSTYLE	164741.0	2019.0
20715	OUTSIDE HARDSTYLE SPED UP	35646.0	329.0
20716	ONLY GIRL HARDSTYLE	6533.0	88.0
20717	MISS YOU HARDSTYLE	158697.0	2484.0

19298 rows × 3 columns

In [111…

```
track_lv['LV_Ratio'] = track_lv['Likes']/track_lv['Views']*100
```

```
C:\Users\hmmuz\AppData\Local\Temp\ipykernel_22064\3962420856.py:1: SettingWithCopyWa
rning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u
ser_guide/indexing.html#returning-a-view-versus-a-copy
  track_lv['LV_Ratio'] = track_lv['Likes']/track_lv['Views']*100
```

In [112…

```
track_lv
```

Out[112…

	Track	Views	Likes	LV_Ratio
0	Feel Good Inc.	693555221.0	6220896.0	0.896958
1	Rhinestone Eyes	72011645.0	1079128.0	1.498547
2	New Gold (feat. Tame Impala and Bootie Brown)	8435055.0	282142.0	3.344874
3	On Melancholy Hill	211754952.0	1788577.0	0.844645
4	Clint Eastwood	618480958.0	6197318.0	1.002022
...	...	...	...	...
20713	JUST DANCE HARDSTYLE	71678.0	1113.0	1.552778
20714	SET FIRE TO THE RAIN HARDSTYLE	164741.0	2019.0	1.225560
20715	OUTSIDE HARDSTYLE SPED UP	35646.0	329.0	0.922965
20716	ONLY GIRL HARDSTYLE	6533.0	88.0	1.347008
20717	MISS YOU HARDSTYLE	158697.0	2484.0	1.565247

19298 rows × 4 columns

In [113…

```
track_lv.sort_values(by = 'LV_Ratio', ascending = False).head(7)
```

Out[113…

	Track	Views	Likes	LV_Ratio
19968	Intro	954081.0	237761.0	24.920421
19969	Safety Zone	1952637.0	453910.0	23.246000
19967	Future	1180522.0	250116.0	21.186899
19971	Pandora's Box	1265231.0	253702.0	20.051832
16297	My Universe - Galantis Remix	2067753.0	371437.0	17.963316
15065	Burn It (feat. MAX)	1054438.0	188244.0	17.852543
8105	No.2 (with parkjiyoon)	2050047.0	346440.0	16.899125

8. What is the Correlatiion between Views, Likes, Comments and Stream?

In [115…

```
df_vlcs = data[['Views', 'Likes', 'Comments', 'Stream']]
df_vlcs
```

Out[115…

	Views	Likes	Comments	Stream
0	693555221.0	6220896.0	169907.0	1.040235e+09
1	72011645.0	1079128.0	31003.0	3.100837e+08
2	8435055.0	282142.0	7399.0	6.306347e+07
3	211754952.0	1788577.0	55229.0	4.346636e+08
4	618480958.0	6197318.0	155930.0	6.172597e+08
...	...	...	...	...
20713	71678.0	1113.0	0.0	9.227144e+06
20714	164741.0	2019.0	0.0	1.089818e+07
20715	35646.0	329.0	0.0	6.226110e+06
20716	6533.0	88.0	0.0	6.873961e+06
20717	158697.0	2484.0	0.0	5.695584e+06

19298 rows × 4 columns

In [116…

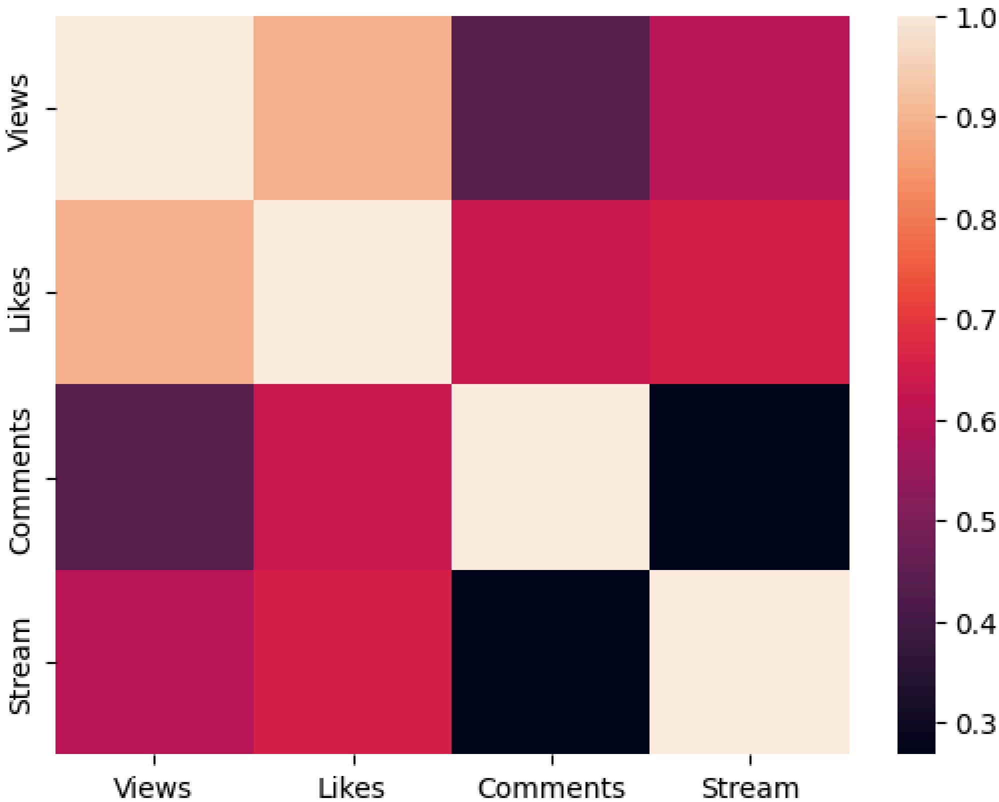
```
df_vlcs.corr()
```

Out[116…

	Views	Likes	Comments	Stream
Views	1.000000	0.891695	0.431077	0.603510
Likes	0.891695	1.000000	0.631035	0.655808
Comments	0.431077	0.631035	1.000000	0.267833
Stream	0.603510	0.655808	0.267833	1.000000

In [118…

```
sns.heatmap(df_vlcs.corr());
```


Views
Likes
Comments
Stream
Views
Likes
Comments
Stream
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3

```
In [ ]:
```